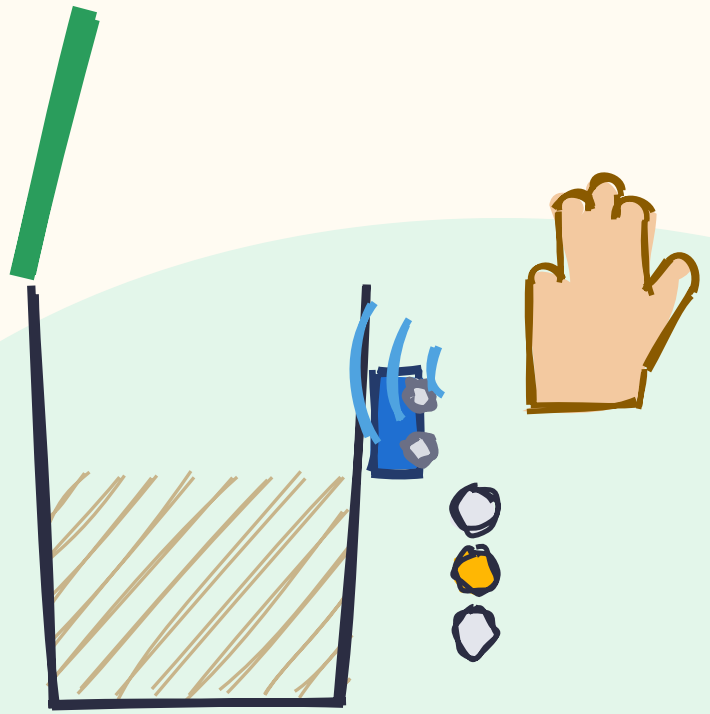
Junior Makers · Class 4

Smart Dustbin

with Fill-Level Indicator

No touching. No overflowing. Just wave!

Ultrasonic sensing | Servo motion | Two sensors, one brain



Did you know?
Bats find insects in total darkness using sound. Our bin uses the same trick!

Let's think...

What is wrong with this bin?

Two big problems

1. Everyone touches the same dirty lid. Germs spread from hand to hand.

2. Nobody knows it is full until rubbish falls on the floor.



overflowing!

Open it with your foot?

Check it every hour?

Make the bin smart!

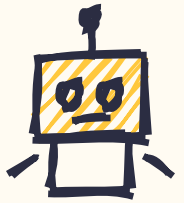
Today: we build a bin that opens when it sees your hand and shows how full it is.

Learning aims

Today I will learn to...

1. Explain how an **ultrasonic sensor** measures distance with echoes
2. Turn a **servo** to an exact angle to open and close a lid
3. Use **two sensors** on one Arduino at the same time
4. Show a **fill level** with a green, yellow and red LED bar
5. Explain how smart bins keep **cities clean**

Keywords
ultrasonic · echo · distance · cm · servo · angle · threshold



Watch and think

Video time!



Echolocation

SciToons · about 5 minutes · youtu.be/laeE4icRYp4

Watch for...

1. How do bats find food in the **dark**?
2. What happens to the sound when it hits an object?
3. How does the **time** of the echo tell the distance?

Scan to watch

After the video: Explain to your partner how our ultrasonic sensor is like a bat. Use the word **echo**.

Real world

Smart bins are already here



Smart city bins
Send a message to the council when they are full, so trucks only go where needed



Hospitals
Touch-free bins stop germs spreading between patients



Fewer truck trips
Less fuel, less traffic, less pollution



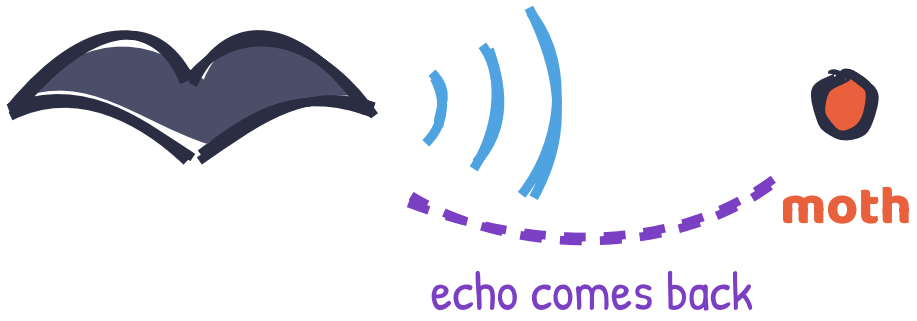
Clean and green
No overflowing bins on the streets or in the school

? Think: what would happen if every bin in Islamabad could tell the cleaners **"I am full, come now"**?

Engineer's eye
The cleverest machines solve small, everyday problems.

The science

Seeing with sound: echoes



A bat shouts a very high squeak. The sound bounces off a moth and comes back. **Quick echo = close. Slow echo = far.**

Ultrasonic means...

Ultra = beyond. **Sonic** = sound.

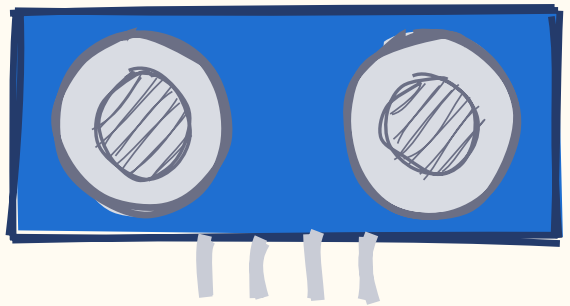
A sound too high for humans to hear. Our sensor sends **40,000 vibrations every second** (40 kHz). We can only hear up to about 20 kHz.

Try it now

Clap once facing the wall, then from far away. Which echo comes back faster?

Meet the sensor

The HC-SR04 ultrasonic sensor



speaker (sends) microphone (hears)

Fact file

Range: **2 cm to 400 cm**

Accuracy: about **3 mm**

Beam: a cone about **15°** wide

Speed of sound: **343 m/s**

Pin	What it does
VCC	5V power
Trig	Arduino says: SHOUT now!
Echo	Sensor says: I heard it back after … microseconds
GND	Ground

The maths inside

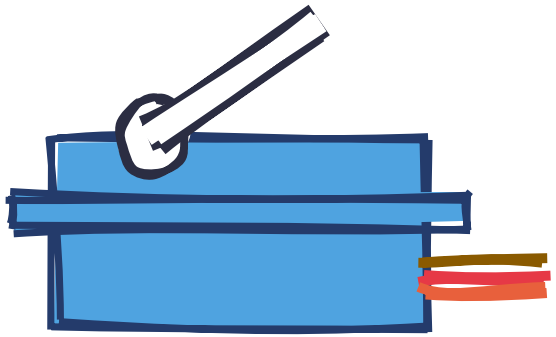
Sound goes there **and** back, so:

distance (cm) = echo time (µs) ÷ 58

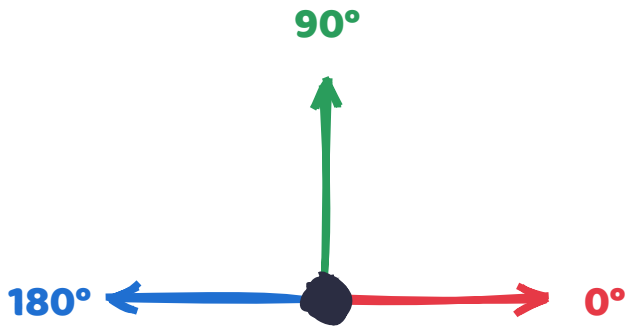
ArduinoBlocks does this maths for you!

Meet the motor

The SG90 servo: a motor that obeys angles



A normal motor spins and spins. A servo turns to an **exact angle** and **holds it**.



It can turn from 0° to 180°

3 wires

Brown → GND

Red → 5V

Orange → Signal: pin 9

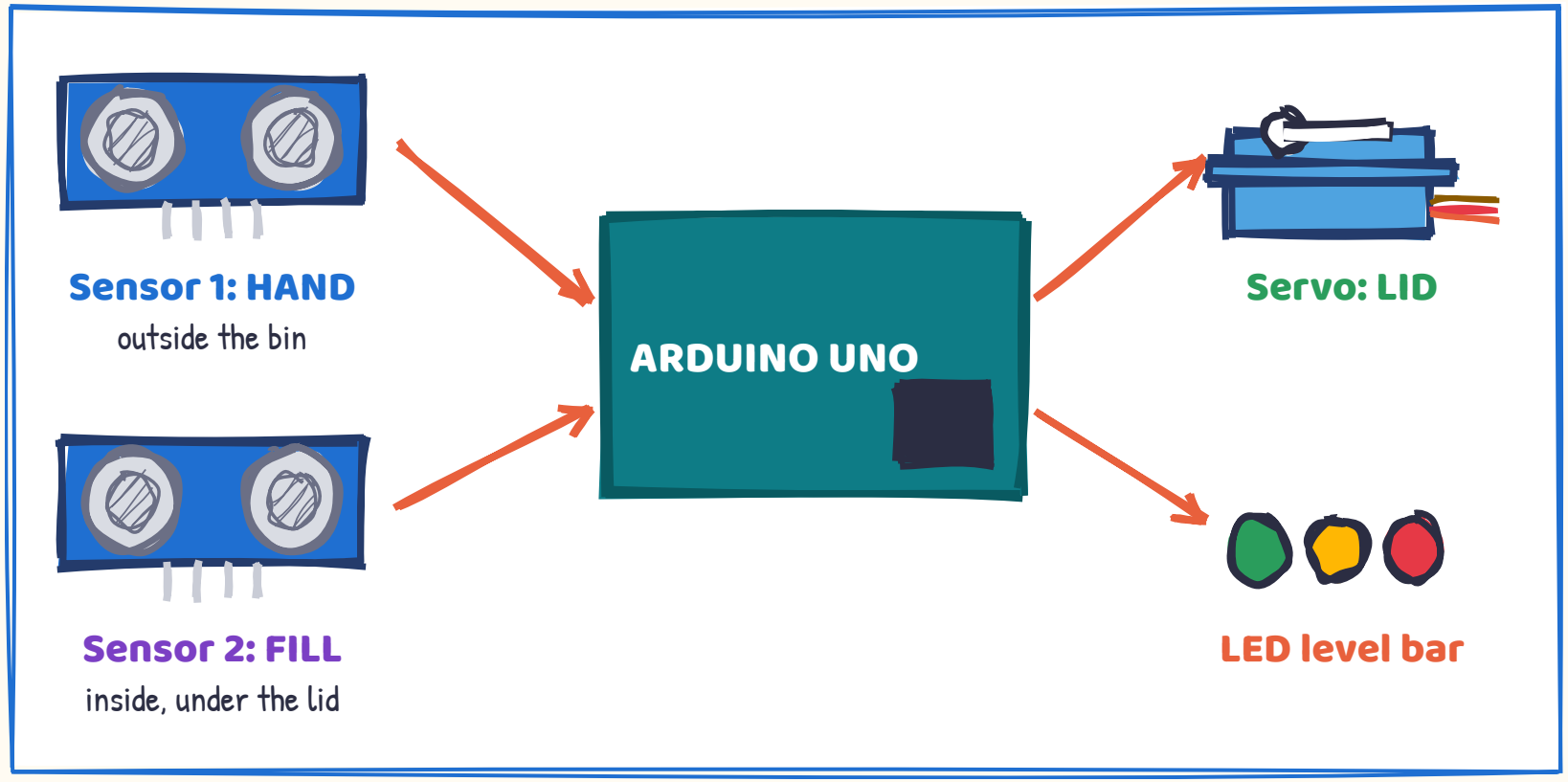
Our lid

0° = lid closed. 90° = lid open. Your bin might need 70° or 110°. Test it!

Inside a servo: a small motor, gears to make it strong, and a sensor that checks the angle. It is a tiny robot on its own!

The plan

Two sensors, one brain, two jobs



Job 1: lid

IF a hand is closer than **15 cm**, open the lid, wait 3 s, close it.

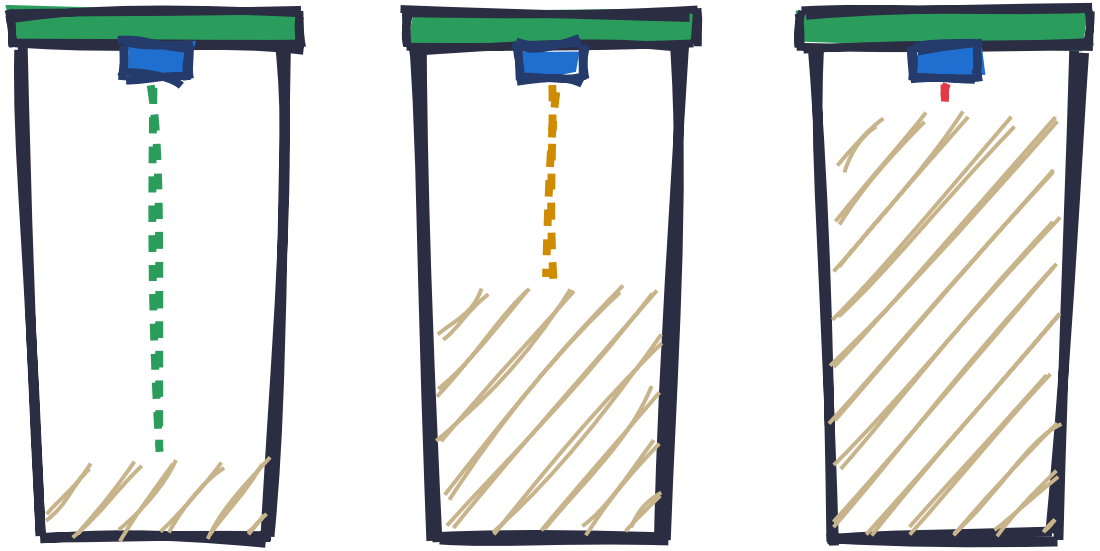
Job 2: level

Measure the gap from the lid down to the rubbish. Show it on the LEDs.

Input → decision → output. Same pattern, now doubled!

The clever part

Measuring how full the bin is



Upside-down thinking!

The sensor looks **down** from the top. More rubbish means a **smaller** distance.
Big number = empty. Small number = full.

LED	Gap to rubbish
Green	more than 14 cm
Yellow	7 to 14 cm
Red	less than 7 cm

For a bin **20 cm deep**. Measure your own box and split it into thirds.

Part 2 · Make it

Now let's put it to work!

You know the ideas. Time to play, build and test. Use these slides while you work:

1. Be the ultrasonic sensor
2. Our smart bin kit (per group)
3. Make the bin body
4. Wiring the smart bin
5. Job 1: the hands-free lid
6. Job 2: the fill-level bar
7. Does every part work?

Stand up and play!

Be the ultrasonic sensor

How to play (pairs)

1. Partner A is the **sensor**, partner B is the **timer**.
2. A rolls a soft ball at the wall and catches it on the way back.
3. B counts how long it takes: "one-elephant, two-elephant..."
4. Move 2 big steps back. Try again!

Distance	Time
Close (1 m)	____ counts
Middle (3 m)	____ counts
Far (5 m)	____ counts

Far away = **longer** time. The sensor does exactly this, but **millions of times faster**.

Think deeper

What if the wall were soft, like a curtain? Would the ball (or the sound) bounce back well?

Build it

Our smart bin kit (per group)

Arduino Uno
The brain

2 × HC-SR04
Hand + fill sensors

SG90 servo
Lifts the lid

3 LEDs + 3 × 220 Ω
Green, yellow, red

Cardboard box
The bin body + lid

Jumper wires
Breadboard + USB

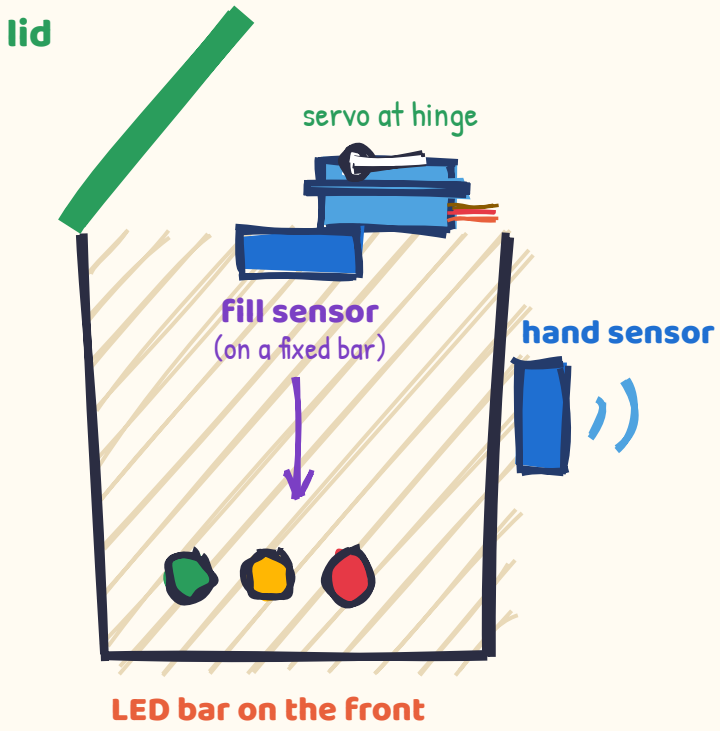
Also: tape, glue gun (teacher only), scissors, a stick or straw for the lid arm, scrap paper "rubbish" to test.

Recycle!
Use an old shoe box or cereal box. Our bin is made from rubbish!

Build it

Make the bin body

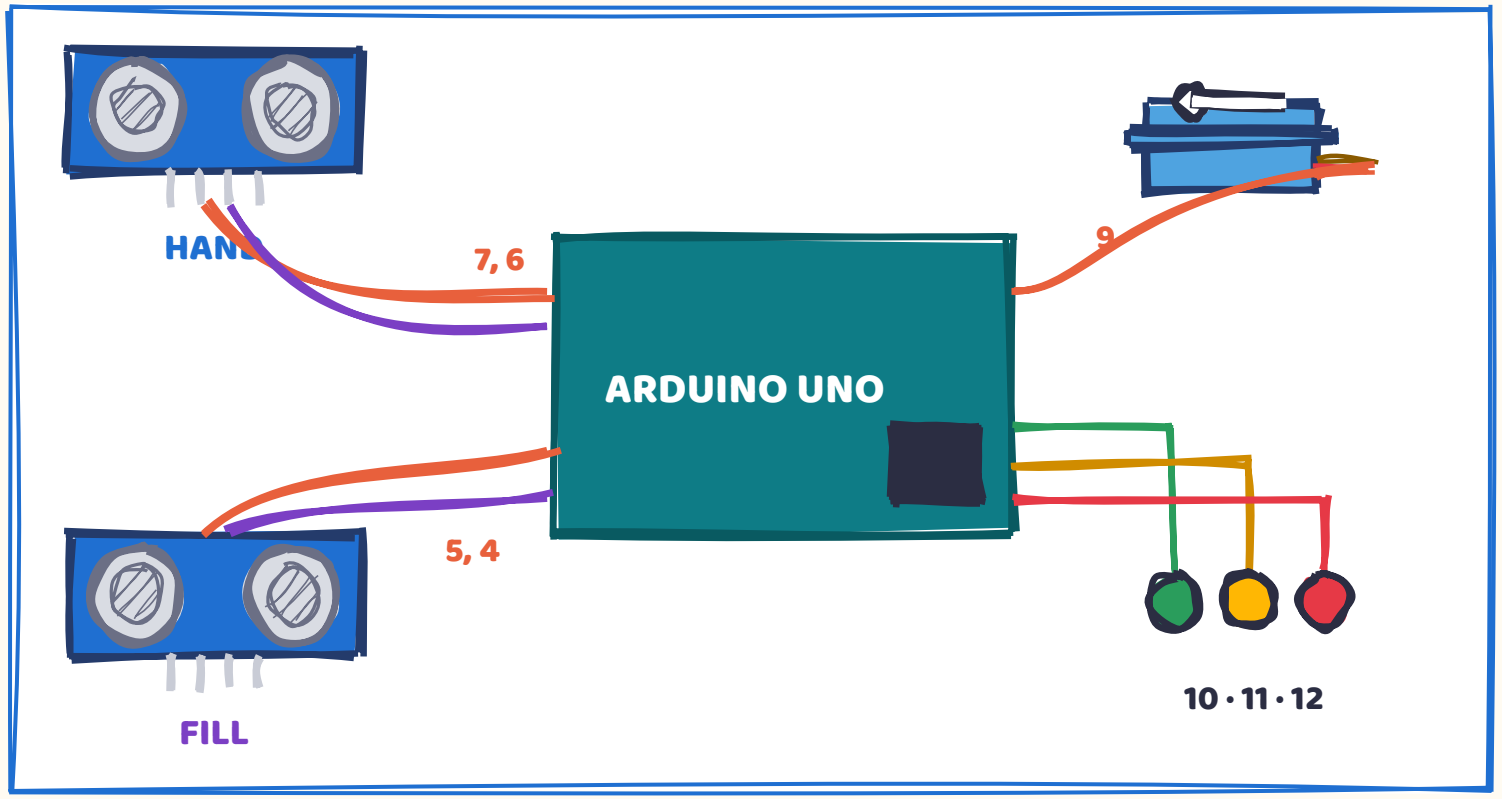
1. **Cut** a flap lid from the top of the box. Leave one side as the hinge.
2. **Fix** the servo next to the hinge. Tape a straw from the horn to the lid.
3. **Mount** the fill sensor on a card bar **under** the lid, looking down. Not on the lid!
4. **Stick** the hand sensor on the front, looking out.
5. **Poke** 3 LEDs through the front: green at the bottom, red at the top.



Why not on the lid? When the lid opens, a sensor on it would point at the ceiling!

Build it

Wiring the smart bin



Part	Arduino
Hand Trig / Echo	7 / 6
Fill Trig / Echo	5 / 4
Sensors VCC / GND	5V / GND
Servo orange	pin 9
Servo red / brown	5V / GND
LEDs G / Y / R	10 / 11 / 12
Each LED	+ 220 Ω

Power tip
Servo twitching or Arduino resetting? Plug into a laptop port directly, or use a 5V adapter.

Code it, part 1

Job 1: the hands-free lid

```
when Arduino starts
    servo pin 9 → angle 0 (lid closed)
forever
    set hand to ultrasonic distance (Trig 7, Echo 6)
    if hand < 15
        servo pin 9 → angle 90 (open)
        wait 3000 ms
        servo pin 9 → angle 0 (close)
        wait 1000 ms
```

Read it like a story

Start with the lid **closed**.
Keep checking for a hand.
IF a hand is closer than 15 cm: open, wait 3 seconds, close.
Wait 1 second so the lid settles.



Common bug
Lid opens by itself? Something is in front of the sensor, like a wall or your bag. Aim it at open space!

Code it, part 2

Job 2: the fill-level bar

```
(still inside forever)
    set fill to ultrasonic distance (Trig 5, Echo 4)
    turn OFF pins 10, 11, 12
    if fill < 7
        turn ON pin 12 (red: FULL)
    else if fill < 14
        turn ON pin 11 (yellow: HALF)
    else
        turn ON pin 10 (green: EMPTY)
    wait 200 ms
```

Ranges, not just yes/no

Last time we used one **IF**. Now we use **IF, ELSE IF, ELSE** to sort a number into **three ranges**.
The order matters: check the smallest number first!

Why one at a time?
Two ultrasonic sensors shouting together can hear each other's echoes. Reading them one after the other keeps them honest.

Test like an engineer

Does every part work?

Test	Expect	Result	Reading
Wave a hand at 10 cm	Lid opens?	pass / fail	_____ cm
Wave at 30 cm	Lid stays shut?	pass / fail	_____ cm
Empty bin	Green?	pass / fail	_____ cm
Half full of paper	Yellow?	pass / fail	_____ cm
Fill to the top	Red?	pass / fail	_____ cm

Debug checklist

Reading stuck at **0**? Check Trig and Echo are not swapped.
Wrong LED? Check the **order** of your IF blocks.
Servo buzzing at 0°? Try **5°** instead.

Open the **Serial monitor** and print both distances. Seeing the numbers makes bugs easy to find!

Level up!

Smart bin challenges

Choose your challenge



Full alarm: add a buzzer that beeps 3 times when the bin turns red.



No entry: when the bin is full, do NOT open the lid. Flash red instead.



Slow close: close the lid gently, a few degrees at a time, with a repeat loop.



Thank you! blink all 3 LEDs as a thank-you after the lid closes.

Design thinking

Who else would love this bin? A doctor? A cook? Someone who cannot bend down? How would you change it for them?

Boss level

Make the lid open only if a hand stays there for half a second. No more accidental opens!

Show what you know

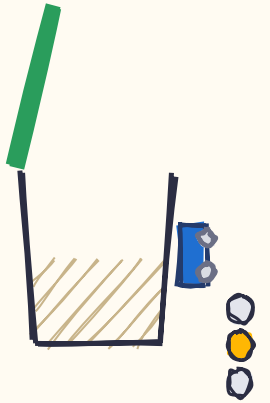
What did we learn?

- [x] An ultrasonic sensor measures distance using echoes, like a bat
- [x] Distance (cm) = echo time (µs) ÷ 58
- [x] A servo turns to an exact angle and holds it
- [x] IF, ELSE IF, ELSE sorts a number into ranges
- [x] Two sensors + one brain = one smart system

Exit ticket: the fill sensor reads 5 cm. Which LED is on, and why?

At home

Find a machine that opens or switches on by itself when you come near. What sensor might it use?



Next class: a Rain-Sensing Automatic Window!